

CWC 2.0 ESM — Codebase Map

A directory-by-directory guide to the ESM service-management module, backend and frontend. All paths relative to repo root.

1. Backend core (backend/src/)

```
backend/src/  
■■■ routes/ Express routers (~46 files) – index.ts is the aggregator  
■■■ controllers/ ~45 HTTP controllers (thin)  
■■■ services/ ~70 service files (business logic)  
■■■ middleware/ auth, RBAC, upload, rate-limit, validate, authorizeResource, etc.  
■■■ utils/ prisma, logger, workflowTransitions (fallback map), redis, sse  
■■■ validators/ Joi request validators  
■■■ constants/ enums/constants (requestStatusCompat, etc.)  
■■■ config/ centralized env config (backend/src/config/index.ts)  
■■■ jobs/, workers/, queues/ scheduled jobs + BullMQ workers  
■■■ templates/ email/PDF templates  
■■■ security/ operation-control registry, etc.
```

Routes — the entry points

Everything mounts in backend/src/routes/index.ts under /api/v1:

URL prefix	Domain
/auth`	auth (before global rate limiter)
/users`	users
/requests`	**request lifecycle** (create/get/update/transitions/approvals/bulk/export)
/service-desks`	**service desks / categories / request types** (public + admin)
/approvals`, /approval-policies`, /approval-delegations`	approvals, policies, delegation
/admin/workflows`, /admin/workflow-versions`, /admin/workflow-transitions`, /admin/status-definitions`	workflow-designer admin
/it-workflow`, /finance-workflow`, /chargeback-workflow`, /esm-workflow`	IT / Finance / chargeback / ESM-travel workflows
/onboarding`, /offboarding`, /loa`, /interviews`, /screening`	HR workflows (+ templates)
/sla`	SLA / escalation rules
/notifications`	notifications (SSE + REST)
/assets`	IT asset management
/kb`	knowledge base
/search`, /reports`, /insights`	search, reports, insights
/files`	file upload/download
/announcements`, /admin/banner-configs`, /admin/system-settings`, /admin/audit-logs`, /admin/tenants`, /admin/entities`, /admin/scheduler`, /admin/queues`	announcements + admin
/esm-workflow`	ESM travel request workflow

Controllers (backend/src/controllers/)

request.controller.ts (the big one), serviceDesk.controller.ts, workflow.controller.ts, workflowVersion.controller.ts, workflowTransition.controller.ts, approval.controller.ts, escalationRule.controller.ts, notification.controller.ts, asset.controller.ts, kb.controller.ts, search.controller.ts, reports.controller.ts, insights.controller.ts, it-workflow.controller.ts, finance-workflow.controller.ts, chargeback-workflow.controller.ts, esm-workflow.controller.ts, onboarding/offboarding/loa/interview/screening.controller.ts, auth.controller.ts, user.controller.ts, plus admin/notification/export/pdf controllers.

Services (backend/src/services/) — key files for orientation

- **Request lifecycle:** requestTransition.service.ts (central transition service — **read first**), availableTransitions.service.ts, requestAccess.service.ts, requestCreationPolicy.service.ts, autoAssignment.service.ts, reassign.service.ts.
- **Workflow engine:** workflowCompiler.service.ts, workflowGraph.service.ts, workflowGraph.types.ts, workflowVersion.service.ts, workflowBootstrap.service.ts, workflowValidator.service.ts, workflowCommand.service.ts, transitionGuards.ts, transitionPolicy.service.ts, conditionalRules.service.ts, conditionEvaluator.service.ts, requestStatusDefinition.service.ts, statusRemap.service.ts.
- **Approvals:** approvalRuntime.service.ts, approvalPolicy.service.ts, approvalDelegation.service.ts.
- **SLA:** sla.service.ts, sla-pause.service.ts.
- **Notifications:** notification.service.ts, outboxDispatcher.service.ts, email.service.ts.
- **Other:** asset / kb / search / insights / reports / announcement / systemSetting / tenant / permission / password-reset / pdfJob / s3 / retentionPolicy.

The two transition systems (IMPORTANT)

- **Runtime (authoritative):** published workflow graphs + the workflow_transitions DB table, consumed by requestTransition.service.ts.
- **Fallback/documentation:** backend/src/utils/workflowTransitions.ts — VALID_TRANSITIONS map of status → allowed next statuses. **Not the runtime source of truth.**

2. Frontend (frontend/)

```
frontend/  
■■■ App.tsx React Router — ESM + CRM + Credit routes  
■■■ pages/ top-level pages (Dashboard, ITSupport, HRServices, GroupFinance, ...)  
■■■ pages/credit/ credit module (separate)  
■■■ src/components/ shared + domain components (FormBuilder, EntityApprovalsPanel, ...)  
■■■ src/services/ per-domain API clients (request.service.ts, serviceDesk.service.ts, ...)  
■■■ src/context/, hooks/, lib/ Auth, notifications, feature-flags, api client  
■■■ src/components/layout/ LeftRail, TopBar, navConfig (buildNavLinks)  
■■■ src/i18n/ global i18n (only Dashboard uses it; ESM is otherwise hard-coded English)
```

Frontend routes (App.tsx, lines 250–270) — ESM core

/ (Dashboard), /hr (HRServices), /it (ITSupport), /finance (GroupFinance), /esm (ExecutiveServices), /my-requests, /request/:id (RequestDetail), /agent (AgentDashboard), /reports (report:read), /insights (report:read), /search, /kb + /kb/:slug (feature-flagged + kb:manage), /approvals

(ApprovalCenter), /inbox (UnifiedInbox), /announcements, /assets (asset:read), plus admin routes. All wrapped in ProtectedRoute.

3. Prisma schema ([backend/prisma/schema.prisma](#))

- **210 models** total. ESM core models: Request, ServiceDesk, ServiceDeskCategory, RequestType, RequestActivity, RequestParticipant, RequestAttachment, RequestApproval, ApprovalInstance, WorkflowType, WorkflowVersion, WorkflowNode, WorkflowEdge, WorkflowStep, WorkflowFormSchema, WorkflowTransition, RequestStatusDefinition, ITHardwareRequest, HRLeaveRequest, FinanceExpenseReimbursement, OnboardingRequest, OffboardingRequest, Candidate, InterviewSchedule, LetterOfAcceptance, EscalationRule, Notification, Asset, AssetAssignment, KbArticle, etc.
- RequestStatus **enum**: ~**110 statuses** across generic / HR-recruitment / IT-procurement / Finance / onboarding / offboarding / inter-company chargeback. RequestStatusLifecycleType (OPEN / RESOLVED / CLOSED / CANCELLED) classifies lifecycle.
- God-models: Request (~40 fields / 40+ relations), User (~163 fields / ~110 relations).

See [\[02-data-model.md\]](#)(02-data-model.md).

4. How to navigate a feature end-to-end (worked example)

Take "an employee submits an IT hardware request and an IT agent progresses it":

- 1 **Frontend** — /it (ITSupport) → create request form (CreateRequest + FormBuilder) calls frontend/src/services/request.service.ts (or it-workflow.service.ts).
- 2 **HTTP** — POST /api/v1/requests (mounted in routes/index.ts) → request.controller.ts::createRequest.
- 3 **Service** — requestCreationPolicy.service.ts + requestTransition.service.ts validate and create the Request (form snapshot, reference number, initial status).
- 4 **Runtime** — the workflow engine (requestTransition.service.ts + published workflow) determines valid next statuses (GET /requests/:id/available-transitions).
- 5 **SLA/notifications** — SLA timer set; in-app + email notifications dispatched (durable outbox).
- 6 **Prisma** — Request, RequestActivity, RequestParticipant, ITHardwareRequest, AssetAssignment.

Follow the same path for any feature: route → controller → service → schema.